CHAPTER 23
Carbonyl Condensation Reactions



FIGURE 23.1 Many of the molecules needed by all growing organisms are biosynthesized using carbonyl condensation reactions. (credit: modification of work "Newborn Sea Lion" by Steven Bedard/Flickr, CC BY 2.0)

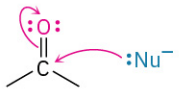
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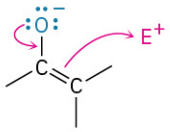
WHY THIS CHAPTER? We'll see later in this chapter and again in Chapter 29 that carbonyl condensation reactions occur in a large number of metabolic pathways. In fact, almost all classes of biomolecules—carbohydrates, lipids, proteins, nucleic acids, and many others—are biosynthesized through pathways that involve carbonyl condensation reactions. As with the α-substitution reaction discussed in the previous chapter, the great value of carbonyl condensations is that they are one of the few general methods for forming carbon–carbon bonds, thereby making it possible to build larger molecules from smaller precursors. In this chapter, we'll see how and why these reactions occur.

We've now studied three of the four general kinds of carbonyl-group reactions and have seen two general

kinds of behavior. In nucleophilic addition and nucleophilic acyl substitution reactions, a carbonyl compound behaves as an electrophile when an electron-rich reagent adds to it. In α-substitution reactions, however, a carbonyl compound behaves as a nucleophile when it is converted into its enol or enolate ion. In the carbonyl condensation reactions that we'll study in this chapter, the carbonyl compound behaves *both* as an electrophile and as a nucleophile.



Electrophilic carbonyl group reacts with nucleophiles.



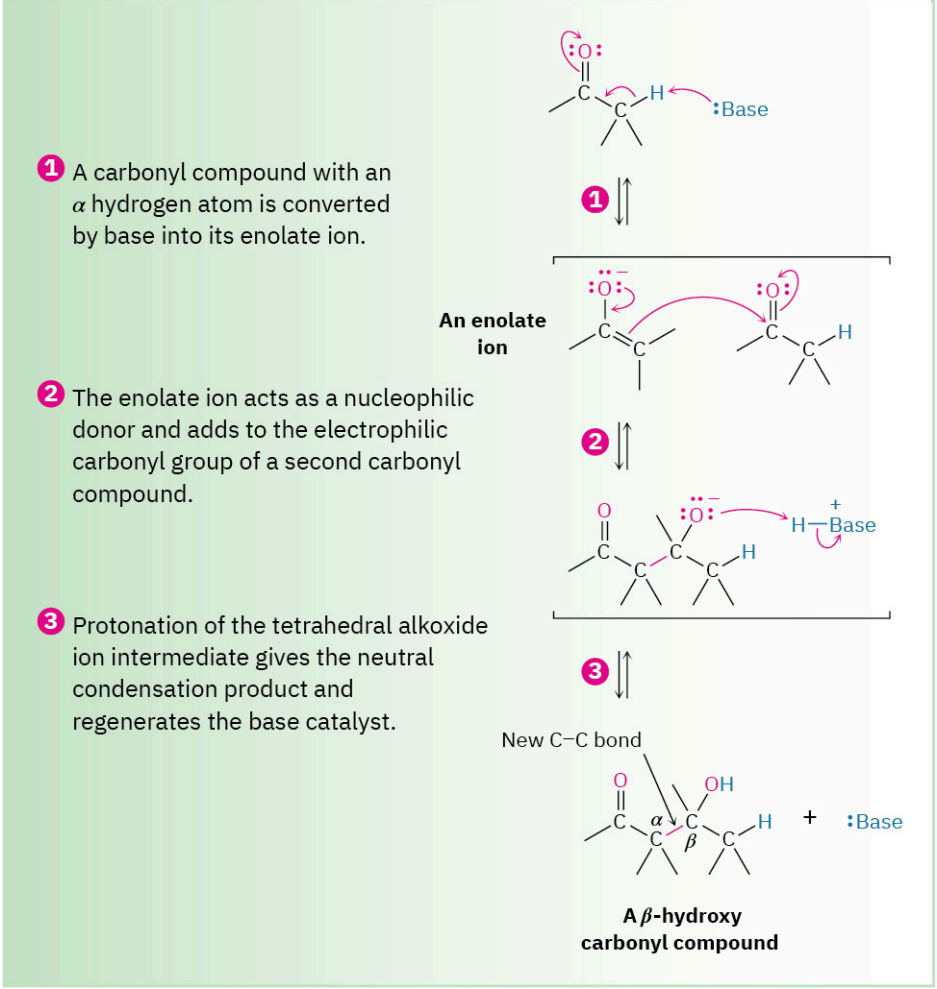
Nucleophilic enolate ion reacts with electrophiles.

23.1 Carbonyl Condensations: The Aldol Reaction

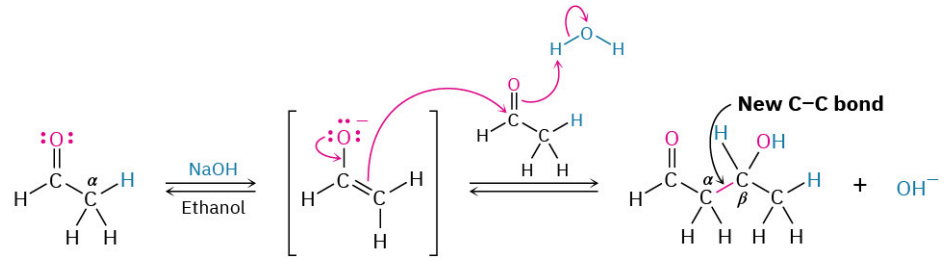
Carbonyl condensation reactions take place between two carbonyl partners and involve a combination of nucleophilic addition and α-substitution steps. One partner is converted into an enolate-ion nucleophile and adds to the electrophilic carbonyl group of the second partner. In so doing, the nucleophilic partner undergoes an α-substitution reaction and the electrophilic partner undergoes a nucleophilic addition. The general mechanism of the process is shown in FIGURE 23.2.

FIGURE 23.2 MECHANISM

The general mechanism of a carbonyl condensation reaction. One partner becomes a nucleophilic donor and adds to the second partner as an electrophilic acceptor. After protonation, the final product is a β-hydroxy carbonyl compound.



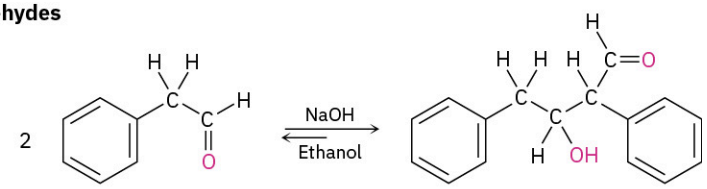
Aldehydes and ketones with an α hydrogen atom undergo a base-catalyzed carbonyl condensation reaction called the **aldol reaction**. For example, treatment of acetaldehyde with a base such as sodium ethoxide or sodium hydroxide in a protic solvent leads to rapid and reversible formation of 3-hydroxybutanal, known commonly as *aldol* (*ald*ehyde + alcoh*ol*), hence the general name of the reaction.



The exact position of the aldol equilibrium depends both on reaction conditions and on substrate structure. The equilibrium generally favors the condensation product in the case of aldehydes with no α substituent

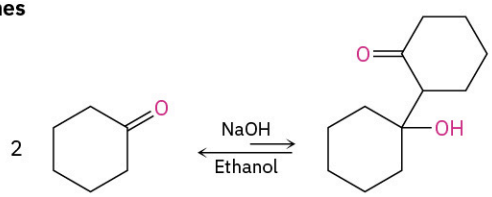
(RCH_2CHO) but favors the reactant for disubstituted aldehydes (R_2CHCHO) and for most ketones. Steric factors are probably responsible for these trends, since increased substitution near the reaction site increases steric congestion in the aldol product.

Aldehydes



Phenylacetaldehyde (10%) **(90%)**

Ketones



Cyclohexanone (78%) **(22%)**

WORKED EXAMPLE 23.1

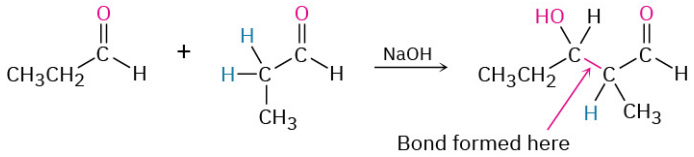
Predicting the Product of an Aldol Reaction

What is the structure of the aldol product from propanal?

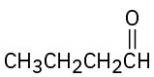
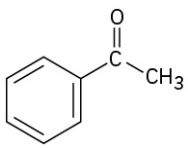
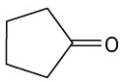
Strategy

An aldol reaction combines two molecules of reactant by forming a bond between the α carbon of one partner and the carbonyl carbon of the second partner. The product is a β-hydroxy aldehyde or ketone, meaning that the two oxygen atoms in the product have a 1,3 relationship.

Solution



PROBLEM 23-1 Predict the aldol reaction product of the following compounds:

(a)  **(b)**  **(c)** 

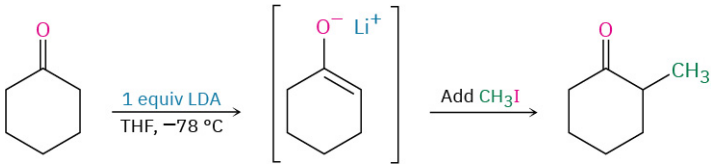
PROBLEM 23-2 Using curved arrows to indicate the electron flow in each step, show how the base-catalyzed reverse-aldol reaction of 4-hydroxy-4-methyl-2-pentanone takes place to yield 2 equivalents of acetone.

23.2 Carbonyl Condensations versus Alpha Substitutions

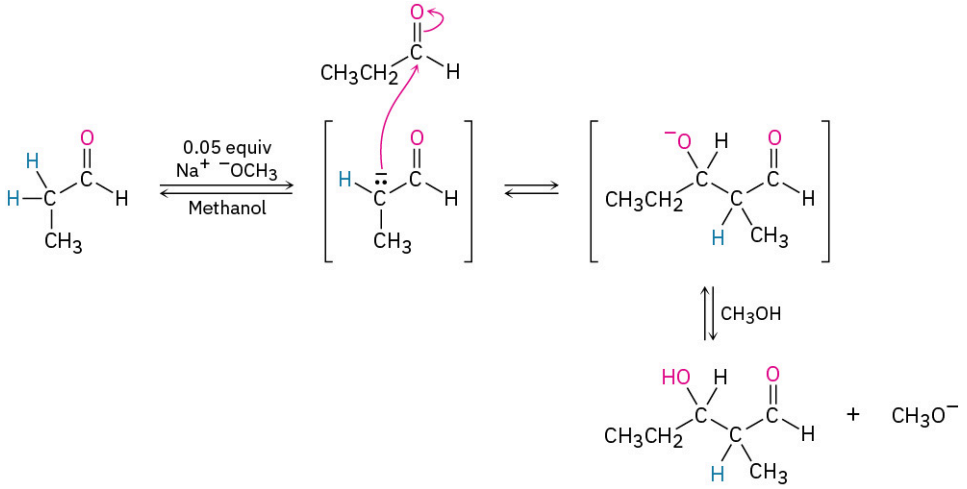
Two of the four general carbonyl-group reactions—carbonyl condensations and α substitutions—take place under basic conditions and involve enolate-ion intermediates. Because the experimental conditions for the two reactions are similar, how can we predict which will occur in a given case? When we generate an enolate ion

with the intention of carrying out an α alkylation, how can we be sure that a carbonyl condensation reaction won't occur instead?

There is no simple answer to this question, but the exact experimental conditions usually have much to do with the result. Alpha-substitution reactions require a full equivalent of strong base and are normally carried out so that the carbonyl compound is rapidly and completely converted into its enolate ion at a low temperature. An electrophile is then added rapidly to ensure that the reactive enolate ion is quenched quickly. In a ketone alkylation reaction, for instance, we might use 1 equivalent of lithium diisopropylamide (LDA) in tetrahydrofuran solution at −78 °C. Rapid and complete generation of the ketone enolate ion would occur, and no unreacted ketone would remain, meaning that no condensation reaction could take place. We would then immediately add an alkyl halide to complete the alkylation reaction.

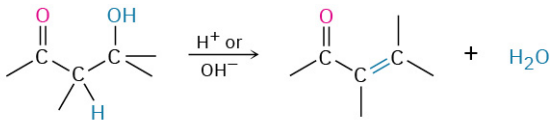


On the other hand, carbonyl condensation reactions require only a catalytic amount of a relatively weak base, rather than a full equivalent, so that a small amount of enolate ion is generated in the presence of unreacted carbonyl compound. Once a condensation occurs, the basic catalyst is regenerated. To carry out an aldol reaction on propanal, for instance, we might dissolve the aldehyde in methanol, add 0.05 equivalent of sodium methoxide, and then warm the mixture to give the aldol product.



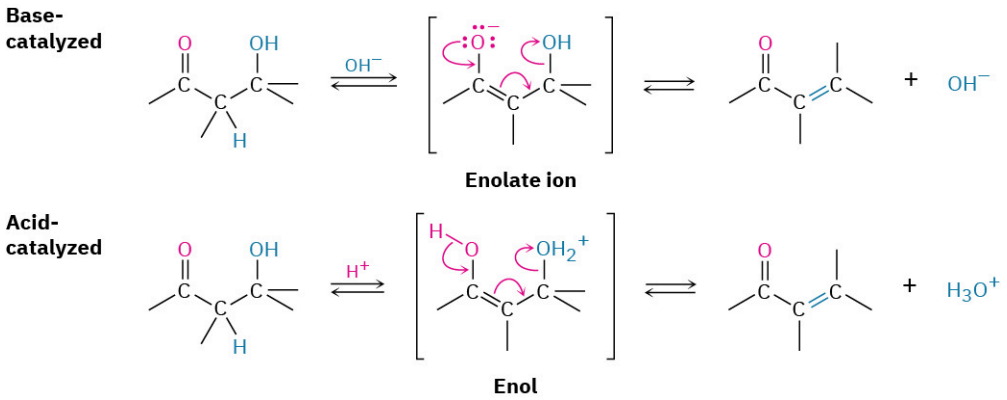
23.3 Dehydration of Aldol Products: Synthesis of Enones

When heated under either acidic or basic conditions, the β-hydroxy aldehydes or ketones formed in aldol reactions can be dehydrated to yield α,β-unsaturated products, or conjugated enones (ene + ketone). In fact, it's this loss of water that gives the carbonyl *condensation* reaction its name, because water condenses out when an enone product forms.



A β-hydroxy ketone or aldehyde **A conjugated enone**

Most alcohols are resistant to dehydration by base (**Section 17.6**) because hydroxide ion is a poor leaving group, but aldol products dehydrate easily because of their carbonyl group. Under basic conditions, an acidic α hydrogen is removed, yielding an enolate ion that expels the nearby $^-$OH leaving group in an E1cB reaction (**Section 11.10**). Under acidic conditions, an enol is formed, the –OH group is protonated, and water is expelled in an E1 or E2 reaction.



The reaction conditions needed for aldol dehydration are often only a bit more vigorous (slightly higher temperature, for instance) than the conditions needed for the aldol formation itself. As a result, conjugated enones are usually obtained directly from aldol reactions without isolating the intermediate β-hydroxy carbonyl compounds.

Conjugated enones are more stable than nonconjugated enones for the same reason that conjugated dienes are more stable than nonconjugated dienes (**Section 14.1**). Interaction between the π electrons of the C═C bond and the π electrons of the C═O group leads to a molecular-orbital description for a conjugated enone with an interaction of the π electrons over all four atomic centers (**FIGURE 23.3**).

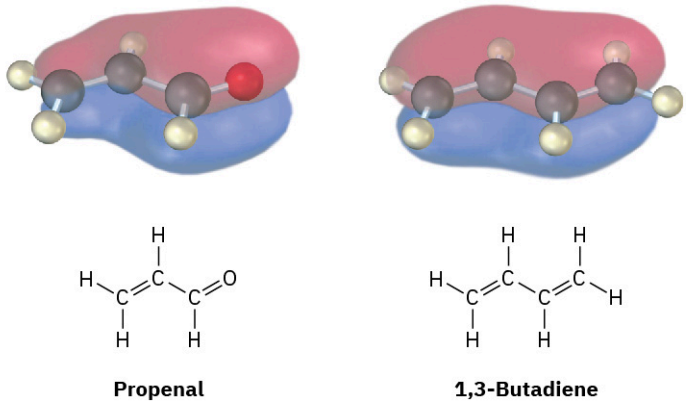
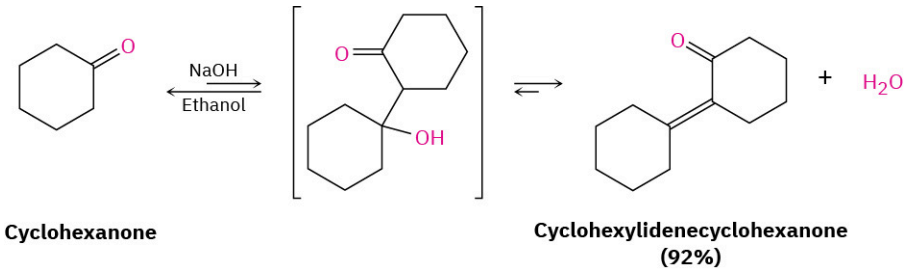


FIGURE 23.3 The π bonding molecular orbitals of a conjugated enone (propenal) and a conjugated diene (1,3-butadiene) are similar in shape and are spread over the entire π system.

The real value of aldol dehydration is that removal of water from the reaction mixture can be used to drive the aldol equilibrium toward the product. Even though the initial aldol step itself may be unfavorable, as it usually is for ketones, the subsequent dehydration step nevertheless allows many aldol condensations to be carried out in good yield. Cyclohexanone, for example, gives cyclohexylidenecyclohexanone in 92% yield, even though the initial equilibrium is unfavorable.



WORKED EXAMPLE 23.2

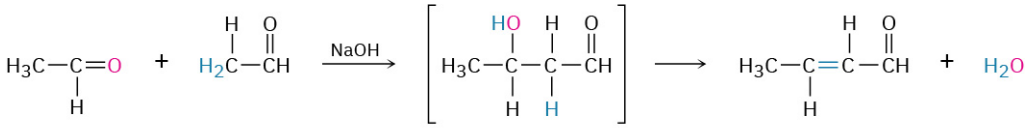
Predicting the Product of an Aldol Reaction

What is the structure of the enone obtained from aldol condensation of acetaldehyde?

Strategy

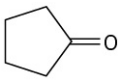
In the aldol reaction, H_2O is eliminated and a double bond is formed by removing two hydrogens from the acidic α position of one partner and the carbonyl oxygen from the second partner. The product is thus an α,β-unsaturated aldehyde or ketone.

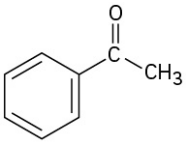
Solution

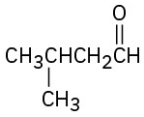


2-Butenal

PROBLEM 23-3 What enone product would you expect from aldol condensation of each of the following compounds?

(a) 

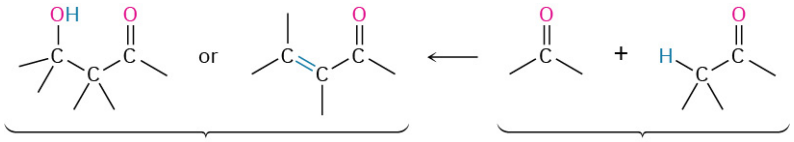
(b) 

(c) 

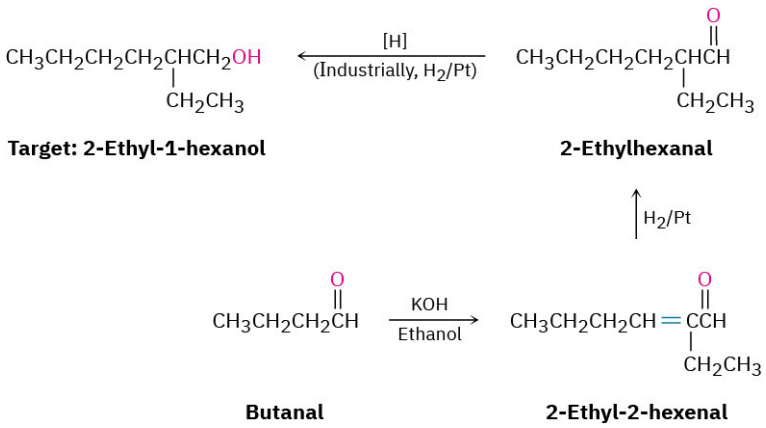
PROBLEM 23-4 Aldol condensation of 3-methylcyclohexanone leads to a mixture of two enone products, not counting double-bond isomers. Draw them.

23.4 Using Aldol Reactions in Synthesis

The aldol reaction yields either a β-hydroxy aldehyde/ketone or an α,β-unsaturated aldehyde/ketone, depending on the experimental conditions. By learning how to work backward, it's possible to predict when the aldol reaction might be useful in synthesis. Whenever the target molecule contains either a β-hydroxy aldehyde/ketone or a conjugated enone functional group, it might come from an aldol reaction.



We can extend this kind of reasoning even further by imagining that subsequent transformations might be carried out on the aldol products. For example, a saturated ketone might be prepared by catalytic hydrogenation of the enone product. A good example can be found in the industrial preparation of 2-ethyl-1-hexanol, an alcohol used in the synthesis of plasticizers for polymers. Although 2-ethyl-1-hexanol bears little resemblance to an aldol product at first glance, it is in fact prepared commercially from butanal by an aldol reaction. Working backward, we can reason that 2-ethyl-1-hexanol might come from 2-ethylhexanal by a reduction. 2-Ethylhexanal, in turn, might be prepared by catalytic reduction of 2-ethyl-2-hexenal, which is the aldol condensation product of butanal. The reactions that follow show the sequence in reverse order.

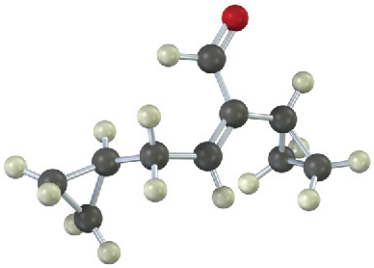


PROBLEM 23-5 Which of the following compounds are aldol condensation products? What is the aldehyde or ketone precursor of each?

(a) 2-Hydroxy-2-methylpentanal **(b)** 5-Ethyl-4-methyl-4-hepten-3-one

PROBLEM 23-6 1-Butanol is prepared commercially by a route that begins with an aldol reaction. Show the steps that are likely to be involved.

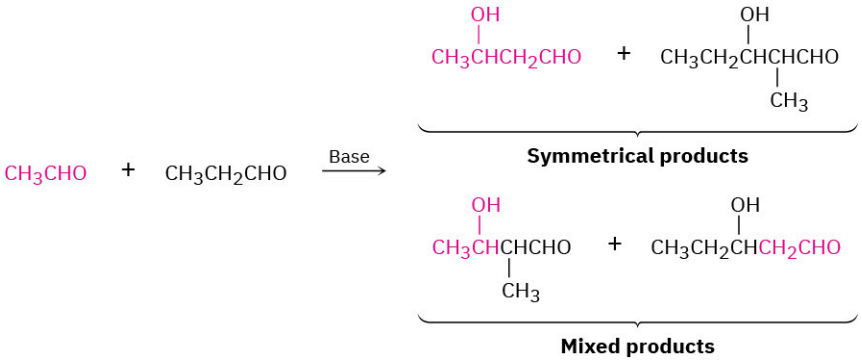
PROBLEM 23-7 Show how you would synthesize the following compound using an aldol reaction:



23.5 Mixed Aldol Reactions

Until now, we've considered only symmetrical aldol reactions, in which the two carbonyl components have been the same. What would happen, though, if an aldol reaction were carried out between two different carbonyl partners?

In general, a mixed aldol reaction between two similar aldehyde or ketone partners leads to a mixture of four possible products. For example, base treatment of a mixture of acetaldehyde and propanal gives a complex product mixture containing two "symmetrical" aldol products and two "mixed" aldol products. Clearly, such a reaction is of no practical value.

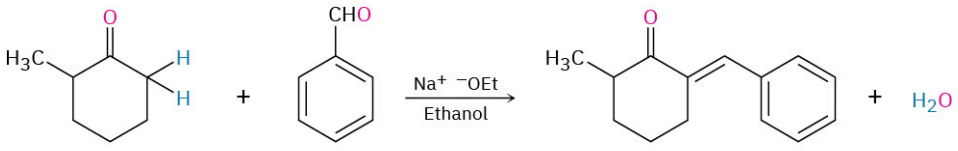


On the other hand, mixed aldol reactions can lead cleanly to a single product if either of two conditions is met:

- If one of the carbonyl partners contains no α hydrogens (and thus can't form an enolate ion to become a donor), but does contain an unhindered carbonyl group (and so is a good acceptor of nucleophiles), then a mixed aldol reaction is likely to be successful. This is the case, for instance, when either benzaldehyde or

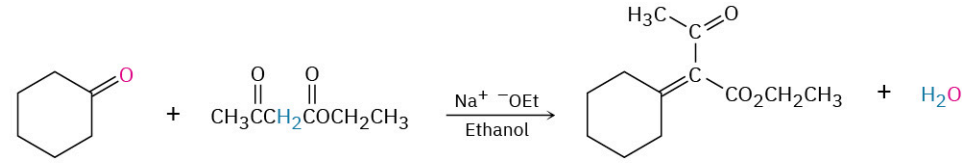
formaldehyde is used as one of the carbonyl partners.

Neither benzaldehyde nor formaldehyde can form an enolate ion to add to another partner, yet both compounds have an unhindered carbonyl group. The ketone 2-methylcyclohexanone, for instance, gives the mixed aldol product on reaction with benzaldehyde.



2-Methylcyclohexanone (donor) **Benzaldehyde (acceptor)** **78%**

- If one of the carbonyl partners is much more acidic than the other, and is thus transformed into its enolate ion in preference to the other, then a mixed aldol reaction is likely to be successful. Ethyl acetoacetate, for instance, is completely converted into its enolate ion in preference to enolate ion formation from monocarbonyl partners. Thus, aldol condensations of monoketones with ethyl acetoacetate occur preferentially to give the mixed product.

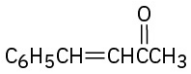


Cyclohexanone (acceptor) **Ethyl acetoacetate (donor)** **80%**

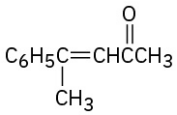
The situation can be summarized by saying that a mixed aldol reaction leads to a mixture of products unless one of the partners either has no α hydrogens but is a good electrophilic acceptor (such as benzaldehyde) or is an unusually acidic nucleophilic donor (such as ethyl acetoacetate).

PROBLEM 23-8 Which of the following compounds can probably be prepared by a mixed aldol reaction? Show the reactants you would use in each case.

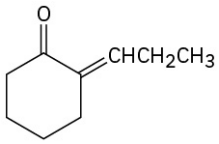
(a)



(b)

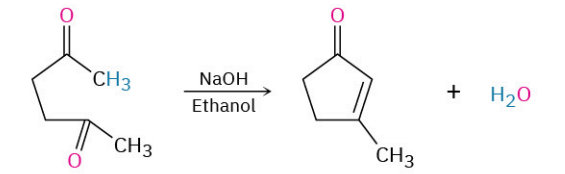


(c)

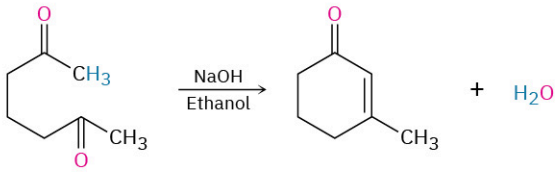


23.6 Intramolecular Aldol Reactions

The aldol reactions we've seen thus far have all been *intermolecular*, meaning that they have taken place between two different molecules. When certain *di*carbonyl compounds are treated with base, however, an *intramolecular* aldol reaction can occur, leading to the formation of a cyclic product. For example, base treatment of a 1,4-diketone such as 2,5-hexanedione yields a cyclopentenone product, and base treatment of a 1,5-diketone such as 2,6-heptanedione yields a cyclohexenone.



2,5-Hexanedione (a 1,4-diketone) **3-Methyl-2-cyclopentenone**



2,6-Heptanedione (a 1,5-diketone) **3-Methyl-2-cyclohexenone**

The mechanism of intramolecular aldol reactions is similar to that of intermolecular reactions. The only difference is that both the nucleophilic carbonyl anion donor and the electrophilic carbonyl acceptor are now in the same molecule. One complication, however, is that intramolecular aldol reactions might lead to a mixture of products, depending on which enolate ion is formed. For example, 2,5-hexanedione might yield either the five-membered-ring product 3-methyl-2-cyclopentenone or the three-membered-ring product (2-methylcyclopropenyl) ethanone (FIGURE 23.4). In practice, though, only the cyclopentenone is formed.

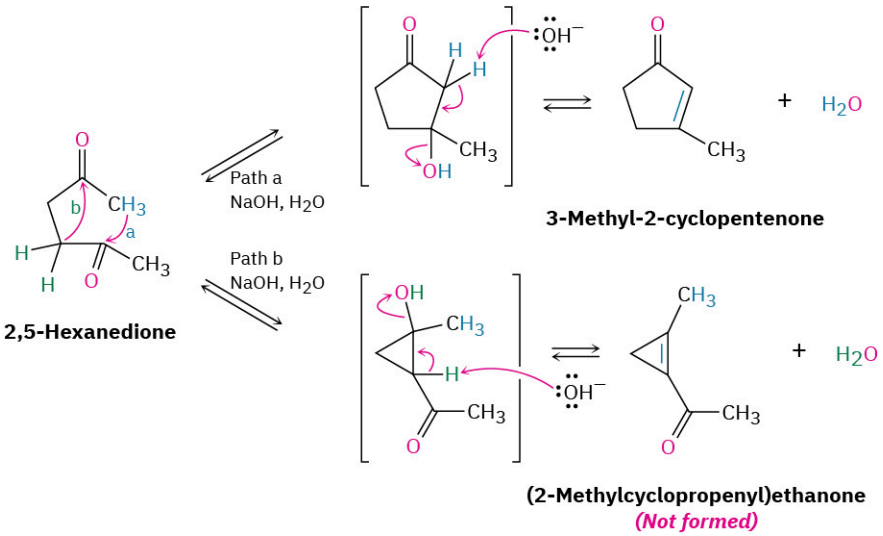
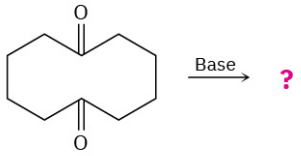


FIGURE 23.4 **Intramolecular aldol reaction of 2,5-hexanedione yields 3-methyl-2-cyclopentenone rather than the alternative cyclopropene.**

The selectivity observed in the intramolecular aldol reaction of 2,5-hexanedione is due to the fact that all steps in the mechanism are reversible, so an equilibrium is reached. Thus, the relatively strain-free cyclopentenone product is considerably more stable than the highly strained cyclopropene alternative. For similar reasons, intramolecular aldol reactions of 1,5-diketones lead only to cyclohexenone products rather than to the more strained acylcyclobutenes.

PROBLEM 23-9 Treatment of a 1,3-diketone such as 2,4-pentanedione with base does not give an aldol condensation product. Explain.

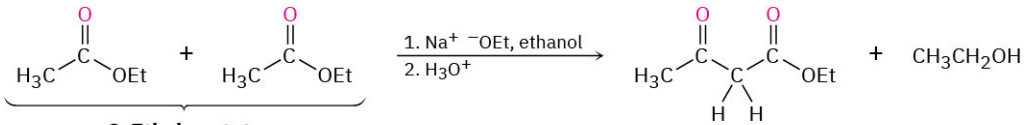
PROBLEM 23-10 What product would you expect to obtain from base treatment of 1,6-cyclodecanedione?



1,6-Cyclodecanedione

23.7 The Claisen Condensation Reaction

Esters, like aldehydes and ketones, are weakly acidic. When an ester with an α hydrogen is treated with 1 equivalent of a base such as sodium ethoxide, a reversible carbonyl condensation reaction occurs to yield a β-keto ester. For instance, ethyl acetate yields ethyl acetoacetate on base treatment. This reaction between two ester molecules is known as the **Claisen condensation reaction**. (We'll use ethyl esters, abbreviated "Et," for consistency, but other esters will also work.)



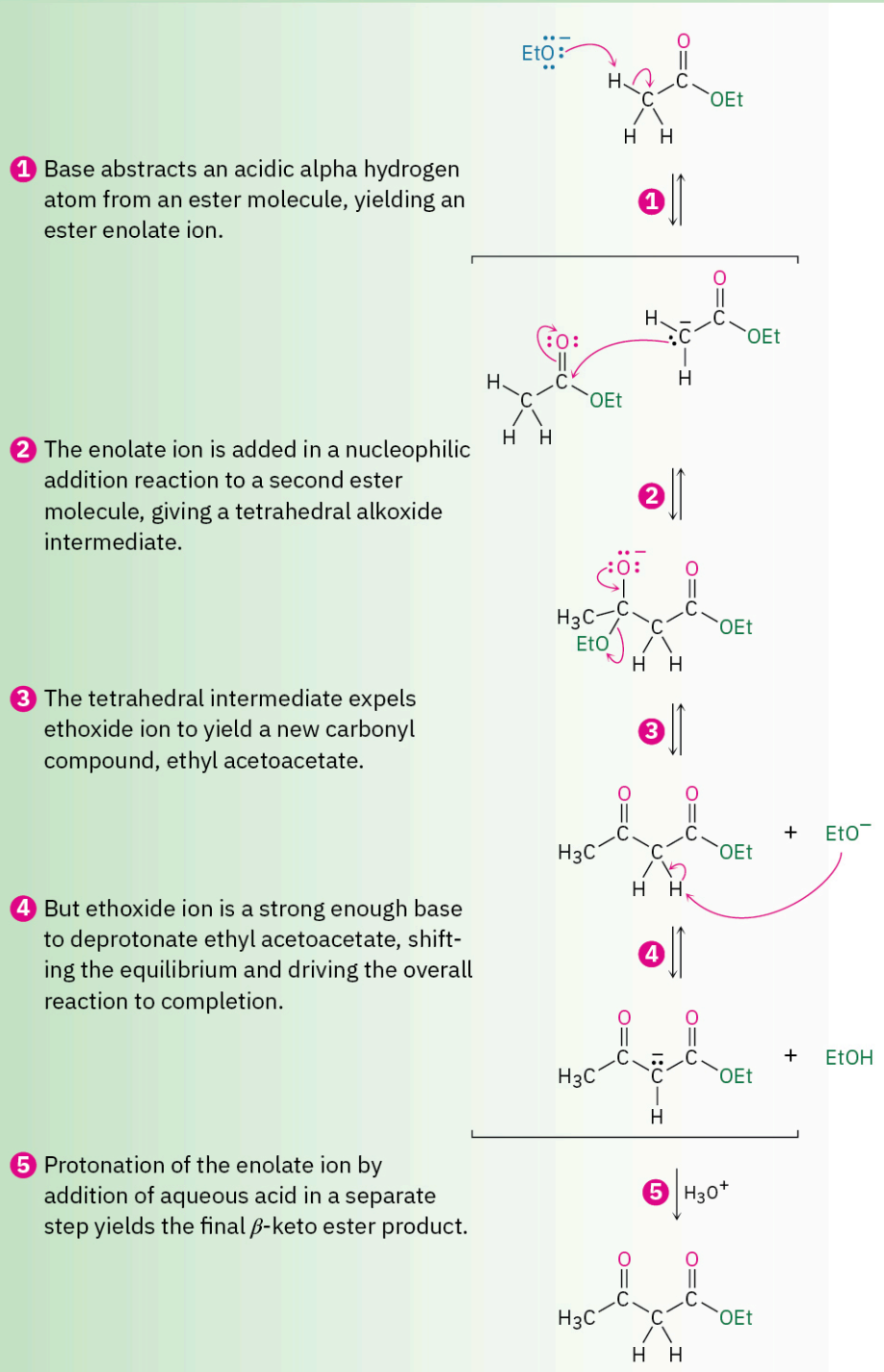
2 Ethyl acetate

Ethyl acetoacetate, a β-keto ester (75%)

The mechanism of the Claisen condensation is similar to that of the aldol condensation and involves the nucleophilic addition of an ester enolate ion to the carbonyl group of a second ester molecule (FIGURE 23.5). The only difference between the aldol condensation of an aldehyde or ketone and the Claisen condensation of an ester involves the fate of the initially formed tetrahedral intermediate. The tetrahedral intermediate in the aldol reaction is protonated to give an alcohol product—exactly the behavior previously seen for aldehydes and ketones (Section 19.4). The tetrahedral intermediate in the Claisen reaction, however, expels an alkoxide leaving group to yield an acyl substitution product—as previously seen for esters (Section 21.6).

FIGURE 23.5 MECHANISM

Mechanism of the Claisen condensation reaction.



If the starting ester has more than one acidic α hydrogen, the product β-keto ester has a highly acidic, doubly activated hydrogen atom that can be abstracted by base. This deprotonation of the product requires the use of a full equivalent of base rather than a catalytic amount. Furthermore, the deprotonation serves to drive the equilibrium completely to the product side so that high yields are usually obtained in Claisen condensations.

WORKED EXAMPLE 23.3

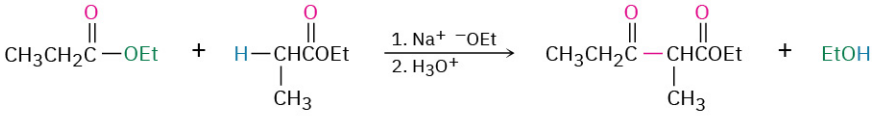
Predicting the Product of a Claisen Condensation Reaction

What product would you obtain from Claisen condensation of ethyl propanoate?

Strategy

The Claisen condensation of an ester results in loss of one molecule of alcohol and formation of a product in which an acyl group of one reactant bonds to the α carbon of the second reactant. The product is a β-keto ester.

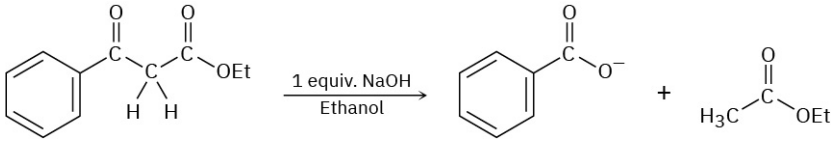
Solution



2 Ethyl propanoate **Ethyl 2-methyl-3-oxopentanoate**

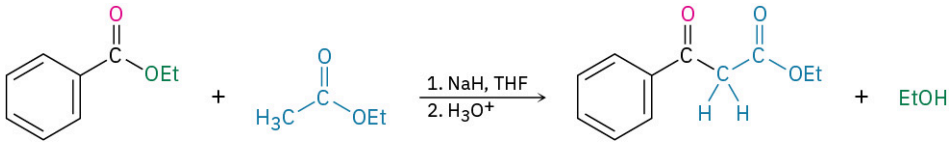
PROBLEM 23-11 Show the products you would expect to obtain by Claisen condensation of the following esters: **(a)** $(CH_3)_2CHCH_2CO_2Et$ **(b)** Ethyl phenylacetate **(c)** Ethyl cyclohexylacetate

PROBLEM 23-12 As shown in Figure 23.5, the Claisen reaction is reversible. That is, a β-keto ester can be cleaved by base into two fragments. Using curved arrows to indicate electron flow, show the mechanism by which this cleavage occurs.



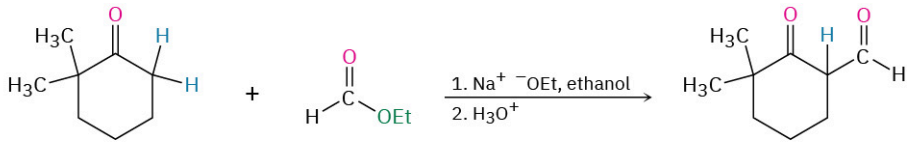
23.8 Mixed Claisen Condensations

The mixed Claisen condensation of two different esters is similar to the mixed aldol condensation of two different aldehydes or ketones (Section 23.5). Mixed Claisen reactions are successful only when one of the two ester components has no α hydrogens and thus can't form an enolate ion. For example, ethyl benzoate and ethyl formate can't form enolate ions and thus can't serve as donors. They can, however, act as the electrophilic acceptor components in reactions with other ester anions to give mixed β-keto ester products.



Ethyl benzoate (acceptor) **Ethyl acetate (donor)** **Ethyl benzoylacetate**

Mixed Claisen-like reactions can also be carried out between an ester and a ketone, resulting in the synthesis of a β-diketone. The reaction works best when the ester component has no α hydrogens and thus can't act as the nucleophilic donor. For example, ethyl formate gives high yields in mixed Claisen condensations with ketones.



2,2-Dimethylcyclohexanone (donor) **Ethyl formate (acceptor)** **A *b*-keto aldehyde (91%)**

WORKED EXAMPLE 23.4

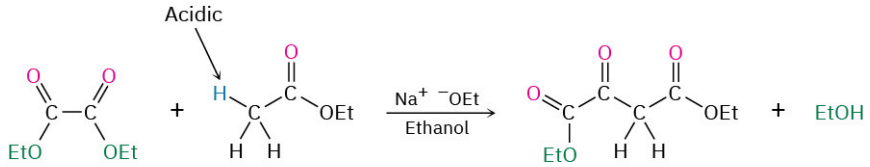
Predicting the Product of a Mixed Claisen Condensation Reaction

Diethyl oxalate, $(CO_2Et)_2$, often gives high yields in mixed Claisen reactions. What product would you expect to obtain from a mixed Claisen reaction of ethyl acetate with diethyl oxalate?

Strategy

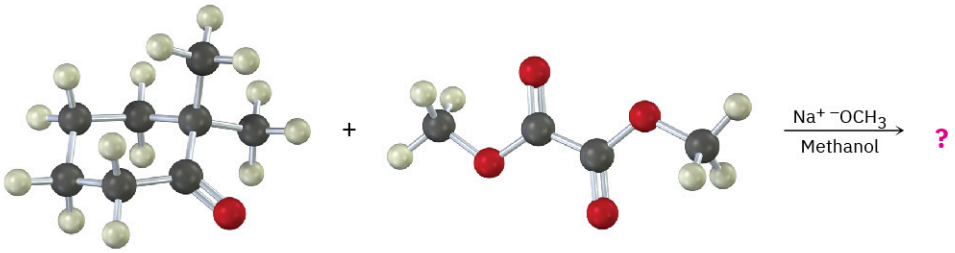
A mixed Claisen reaction is effective when only one of the two partners has an acidic α hydrogen atom. In the present case, ethyl acetate can be converted into its enolate ion, but diethyl oxalate cannot. Thus, ethyl acetate acts as the donor and diethyl oxalate as the acceptor.

Solution



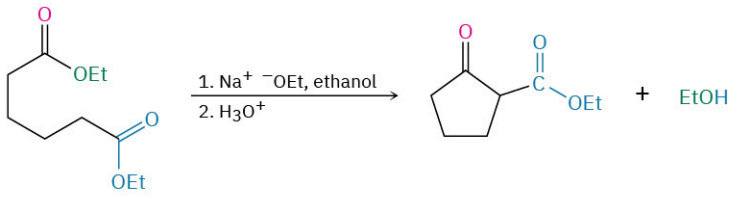
Diethyl oxalate **Ethyl acetate**

PROBLEM 23-13 What product would you expect from the following mixed Claisen-like reaction?

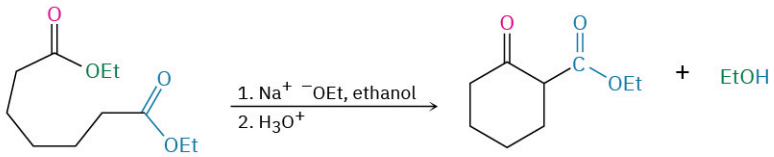


23.9 Intramolecular Claisen Condensations: The Dieckmann Cyclization

Intramolecular Claisen condensations can be carried out with diesters, just as intramolecular aldol condensations can be carried out with diketones (Section 23.6). Called the **Dieckmann cyclization**, this reaction works best on 1,6-diesters and 1,7-diesters. Intramolecular Claisen cyclization of a 1,6-diester gives a five-membered cyclic β-keto ester, and cyclization of a 1,7-diester gives a six-membered cyclic β-keto ester.



Diethyl hexanedioate (a 1,6-diester) **Ethyl 2-oxocyclopentanecarboxylate (82%)**

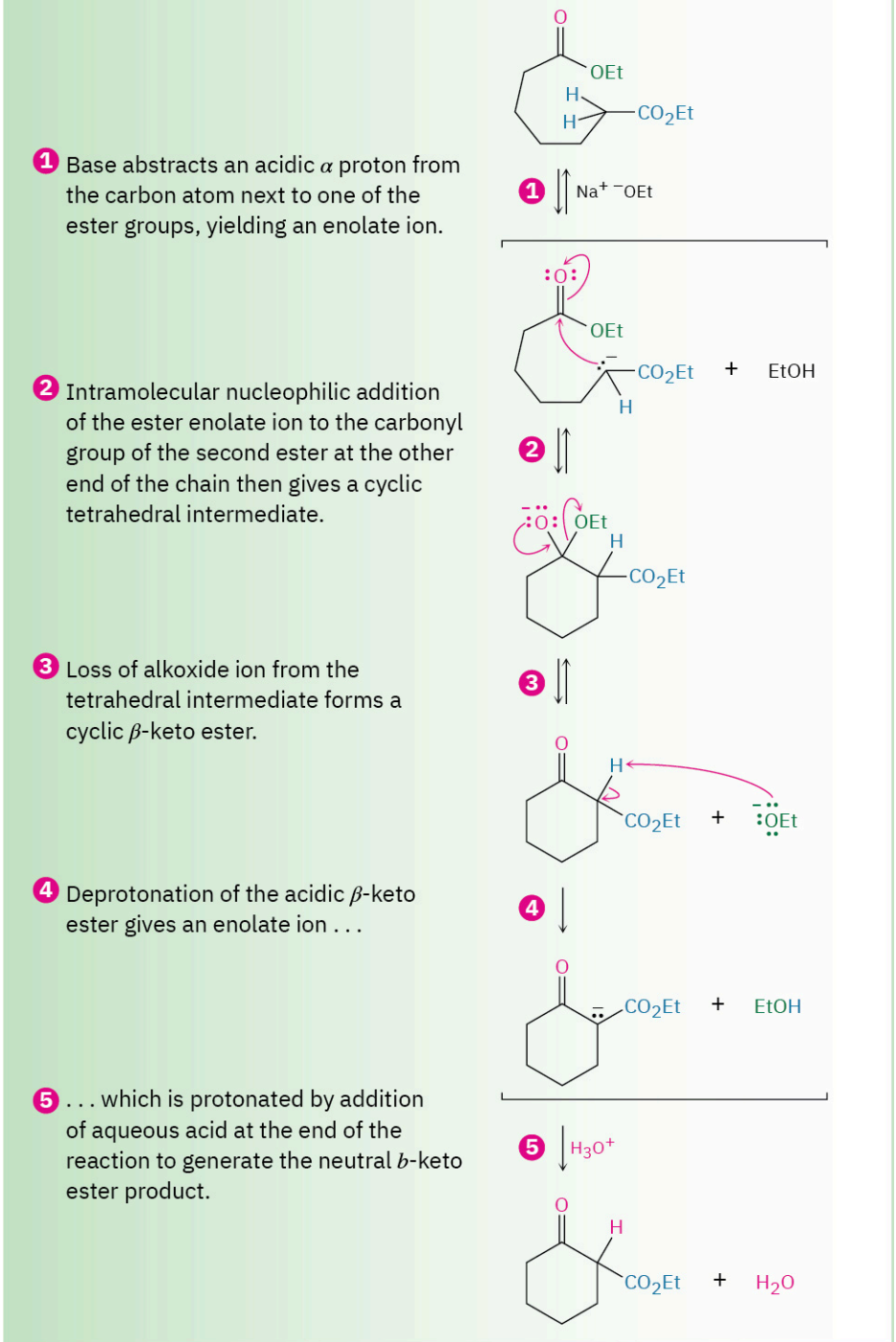


Diethyl heptanedioate (a 1,7-diester) **Ethyl 2-oxocyclohexanecarboxylate**

The mechanism of the Dieckmann cyclization, shown in **FIGURE 23.6**, is the same as that of the Claisen condensation. One of the two ester groups is converted into an enolate ion, which carries out a nucleophilic acyl substitution on the second ester group at the other end of the molecule. A cyclic β-keto ester product results.

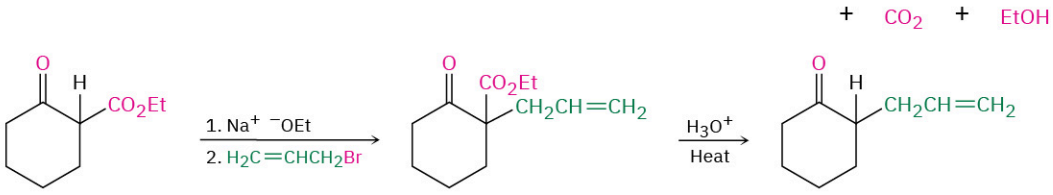
FIGURE 23.6 MECHANISM

Mechanism of the Dieckmann cyclization of a 1,7-diester to yield a cyclic β-keto ester product.



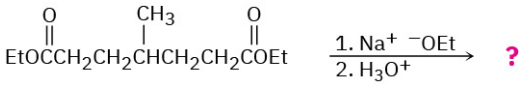
The cyclic β-keto ester produced in a Dieckmann cyclization can be further alkylated and decarboxylated by a series of reactions analogous to those used in the acetoacetic ester synthesis (**Section 22.7**). Alkylation and

subsequent decarboxylation of ethyl 2-oxocyclohexanecarboxylate, for instance, yields a 2-alkylcyclohexanone. The overall sequence of (**1**) Dieckmann cyclization, (**2**) β-keto ester alkylation, and (**3**) decarboxylation is a powerful method for preparing 2-substituted cyclopentanones and cyclohexanones.



Ethyl 2-oxocyclohexanecarboxylate — **2-Allylcyclohexanone (83%)**

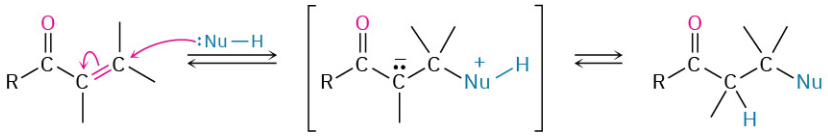
PROBLEM 23-14 What product would you expect from the following reaction?



PROBLEM 23-15 Dieckmann cyclization of diethyl 3-methylheptanedioate gives a mixture of two β-keto ester products. What are their structures, and why is a mixture formed?

23.10 Conjugate Carbonyl Additions: The Michael Reaction

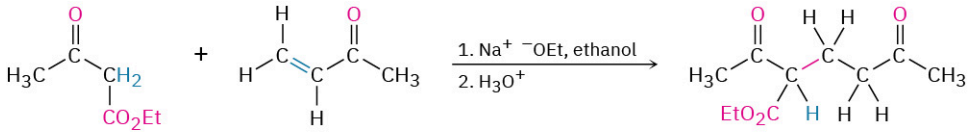
We saw in **Section 19.13** that certain nucleophiles, such as amines, react with α,β-unsaturated aldehydes and ketones to give a conjugate addition product, rather than a direct addition product.



Conjugate addition product

Exactly the same kind of conjugate addition can occur when a nucleophilic enolate ion reacts with an α,β-unsaturated carbonyl compound—a process known as the **Michael reaction** after Arthur Michael at Tufts College and Harvard University.

The best Michael reactions are those that take place when a particularly stable enolate ion, such as that derived from a β-keto ester or other 1,3-dicarbonyl compound, adds to an unhindered α,β-unsaturated ketone. For example, ethyl acetoacetate reacts with 3-buten-2-one in the presence of sodium ethoxide to yield the conjugate addition product.

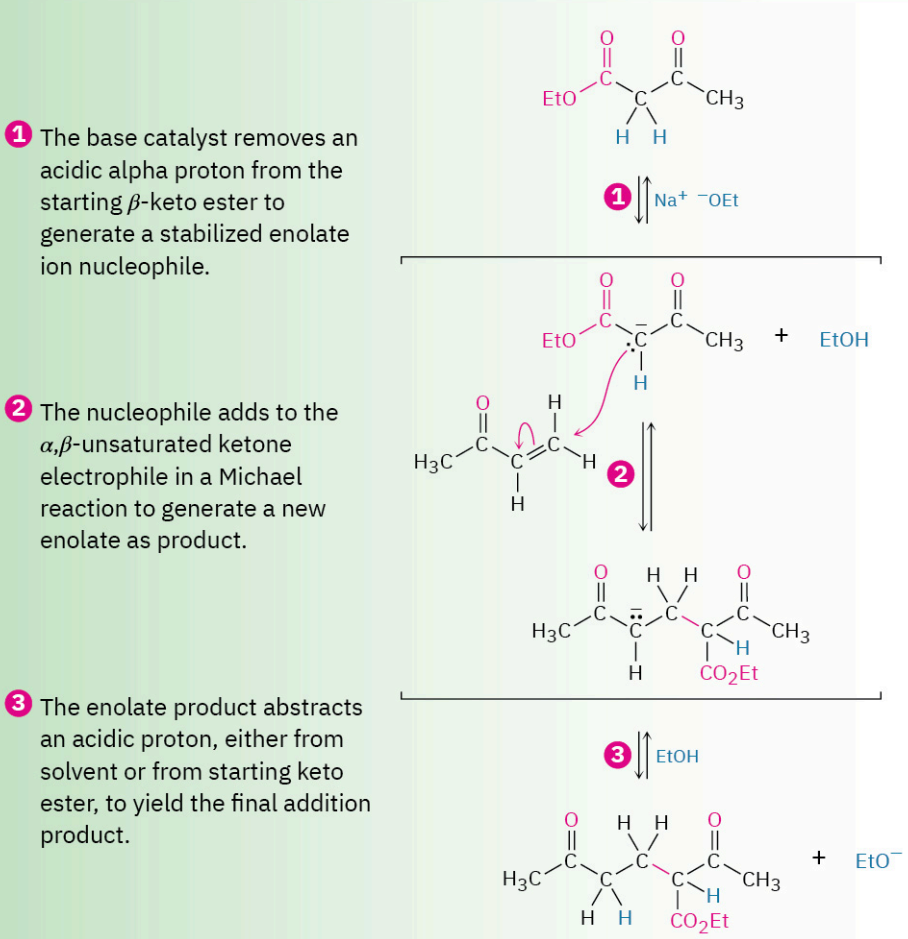


Ethyl acetoacetate — **3-Buten-2-one**

Michael reactions take place by addition of a nucleophilic enolate ion donor to the β carbon of an α,β-unsaturated carbonyl acceptor, according to the mechanism shown in **FIGURE 23.7**.

FIGURE 23.7 MECHANISM

Mechanism of the Michael reaction between a β-keto ester and an α,β-unsaturated ketone. The reaction is a conjugate addition of an enolate ion to the unsaturated carbonyl compound.

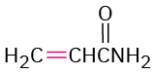
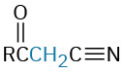
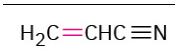
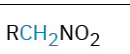
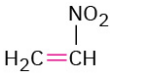


The Michael reaction occurs with a variety of α,β-unsaturated carbonyl compounds, not just conjugated ketones. Unsaturated aldehydes, esters, thioesters, nitriles, amides, and nitro compounds can all act as the electrophilic acceptor component in Michael reactions (TABLE 23.1). Similarly, a variety of different donors can be used, including β-diketones, β-keto esters, malonic esters, β-keto nitriles, and nitro compounds.

TABLE 23.1 Some Michael Acceptors and Michael Donors

Michael acceptors		Michael donors	
$H_2C{=}CHCH{=}O$	Propenal	$RCCH_2CR'$ (O, O)	β-Diketone
$H_2C{=}CHCCH_3$ (O)	3-Buten-2-one	$RCCH_2COEt$ (O, O)	β-Keto ester
$H_2C{=}CHCOEt$ (O)	Ethyl propenoate	$EtOCCH_2COEt$ (O, O)	Diethyl malonate

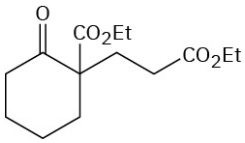
TABLE 23.1 Some Michael Acceptors and Michael Donors

Michael acceptors		Michael donors	
	Propenamide		β-Keto nitrile
	Propenenitrile		Nitro compound
	Nitroethylene		

WORKED EXAMPLE 23.5

Using the Michael Reaction

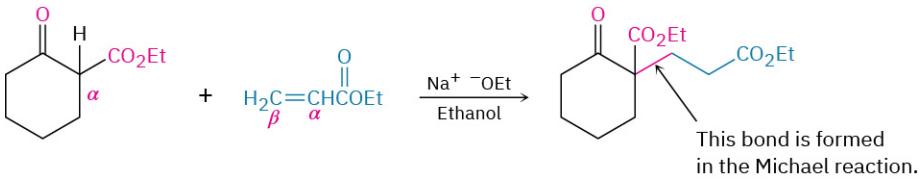
How might you obtain the following compound using a Michael reaction?



Strategy

A Michael reaction involves the conjugate addition of a stable enolate ion donor to an α,β-unsaturated carbonyl acceptor, yielding a 1,5-dicarbonyl product. Usually, the stable enolate ion is derived from a β-diketone, β-keto ester, malonic ester, or similar compound. The C–C bond formed in the conjugate addition step is the one between the α carbon of the acidic donor and the β carbon of the unsaturated acceptor.

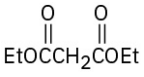
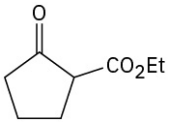
Solution



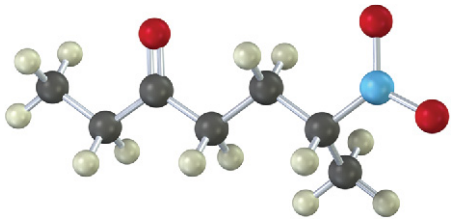
PROBLEM 23-16 What product would you obtain from a base-catalyzed Michael reaction of 2,4-pentanedione with each of the following α,β-unsaturated acceptors?

(a) 2-Cyclohexenone **(b)** Propenenitrile **(c)** Ethyl 2-butenoate

PROBLEM 23-17 What product would you obtain from a base-catalyzed Michael reaction of 3-buten-2-one with each of the following nucleophilic donors?

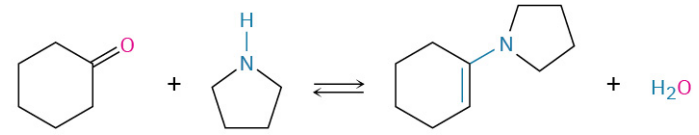
(a)  **(b)** 

PROBLEM 23-18 How would you prepare the following compound using a Michael reaction (gray = H, black = C, red = O, blue = N)?



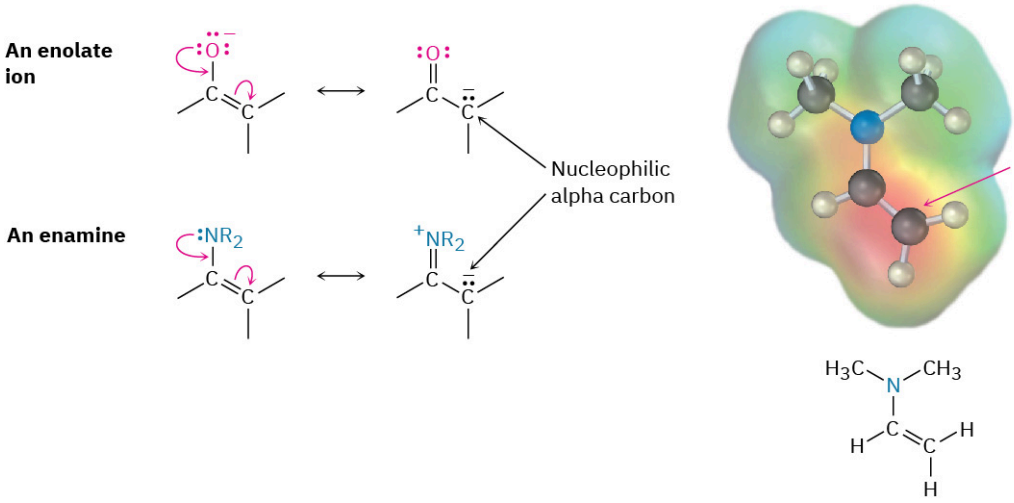
23.11 Carbonyl Condensations with Enamines: The Stork Enamine Reaction

In addition to enolate ions, other kinds of carbon nucleophiles also add to α,β-unsaturated acceptors in Michael-like reactions. Among the most important and useful of such nucleophiles, particularly in biological chemistry, are *enamines,* which are readily prepared by reaction between a ketone and a secondary amine (Section 19.8). For example:



Cyclohexanone **Pyrrolidine** **1-Pyrrolidino-cyclohexene (87%)**

As the following resonance structures indicate, enamines are electronically similar to enolate ions. Overlap of the nitrogen lone-pair orbital with the double-bond *p* orbitals leads to an increase in electron density on the α carbon atom, making that carbon nucleophilic. An electrostatic potential map of *N,N*-dimethylaminoethylene shows this shift of electron density (red) toward the α position.



Enamines behave in much the same way as enolate ions and enter into many of the same kinds of reactions. In the **Stork enamine reaction**, named for its discoverer, Gilbert Stork of Columbia University, for example, an enamine adds to an α,β-unsaturated carbonyl acceptor in a Michael-like process. The initial product is then hydrolyzed by aqueous acid to yield a 1,5-dicarbonyl compound. The overall reaction is thus a three-step sequence of (1) enamine formation from a ketone, (2) Michael addition to an α,β-unsaturated carbonyl compound, and (3) enamine hydrolysis back to a ketone.

The net effect of the Stork enamine reaction is the Michael addition of a ketone to an α,β-unsaturated carbonyl compound. For example, cyclohexanone reacts with the cyclic amine pyrrolidine to yield an enamine; further reaction with an enone such as 3-buten-2-one yields a Michael adduct; and aqueous hydrolysis completes the sequence to give a 1,5-diketone (FIGURE 23.8).

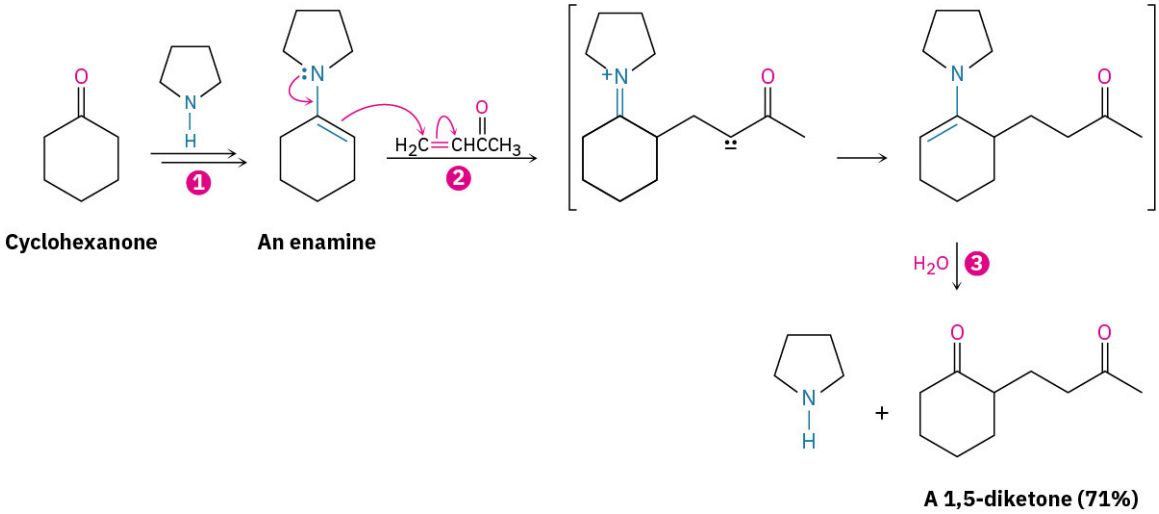


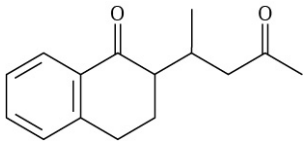
FIGURE 23.8 **The Stork enamine reaction between cyclohexanone and 3-buten-2-one.** (1) Cyclohexanone is first converted into an enamine, (2) the enamine adds to the α,β-unsaturated ketone in a Michael reaction, and (3) the conjugate addition product is hydrolyzed to yield a 1,5-diketone.

The enamine–Michael reaction has two advantages over the enolate-ion–Michael reaction that makes it particularly useful in biological pathways. First, an enamine is neutral, easily prepared, and easily handled, while an enolate ion is charged, sometimes difficult to prepare, and must be handled carefully. Second, an enamine from a *mono*ketone can be used in the Michael addition, whereas only enolate ions from β-dicarbonyl compounds can be used.

WORKED EXAMPLE 23.6

Using the Stork Enamine Reaction

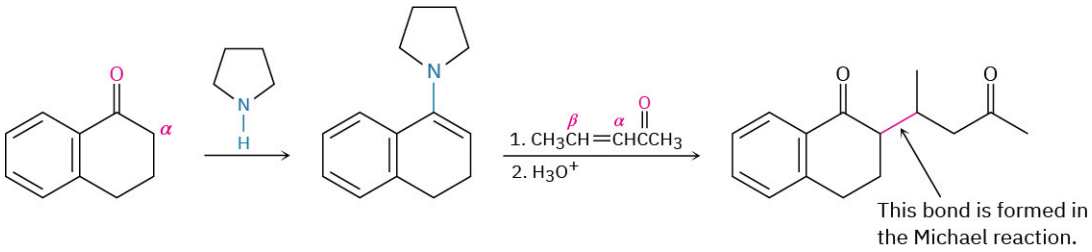
How might you use an enamine reaction to prepare the following compound?



Strategy

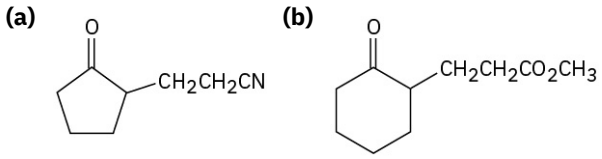
The overall result of an enamine reaction is the Michael addition of a ketone as donor to an α,β-unsaturated carbonyl compound as acceptor, yielding a 1,5-dicarbonyl product. The C–C bond formed in the Michael addition step is the one between the α carbon of the ketone donor and the β carbon of the unsaturated acceptor.

Solution



PROBLEM 23-19 What products would result after hydrolysis from reaction of the enamine prepared from cyclopentanone and pyrrolidine with the following α,β-unsaturated acceptors?
(a) $H_2C{=}CHCO_2Et$ **(b)** $H_2C{=}CHCHO$ **(c)** $CH_3CH{=}CHCOCH_3$

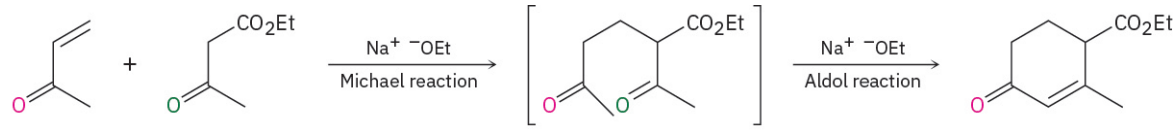
PROBLEM 23-20 Show how you might use an enamine reaction to prepare each of the following compounds:



23.12 The Robinson Annulation Reaction

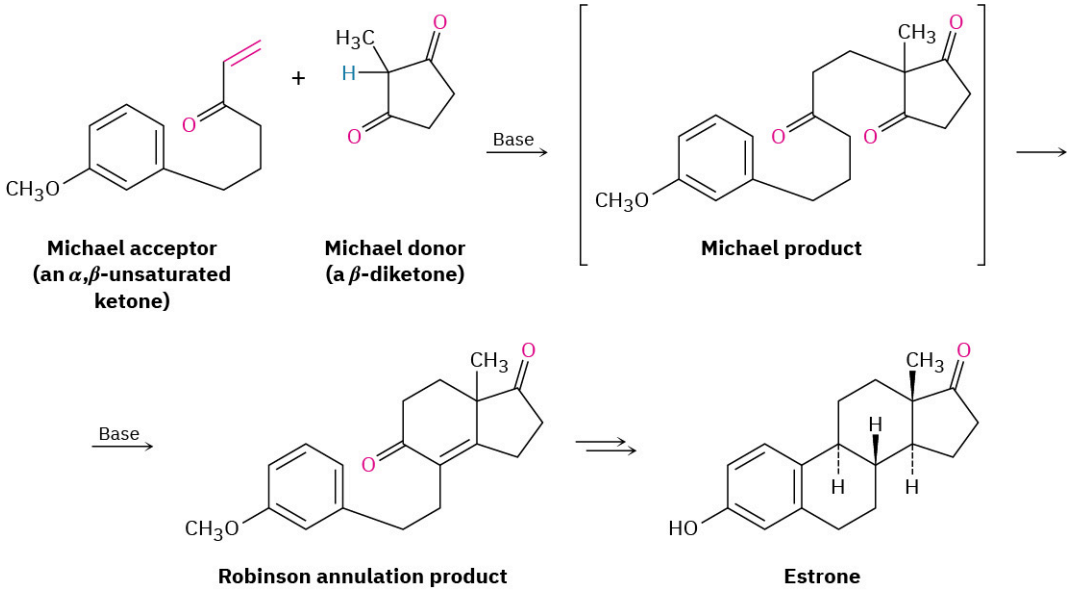
Carbonyl condensation reactions are perhaps the most versatile methods available for synthesizing complex molecules. By putting a few fundamental reactions together in the proper sequence, some remarkably useful transformations can be carried out. One such example is the **Robinson annulation reaction** for the synthesis of polycyclic molecules. The word *annulation* derives from the Latin *annulus,* meaning "ring," so an annulation reaction is one that builds a new ring onto a molecule.

Named after Robert Robinson, a British chemist and 1947 Nobel Prize winner, the Robinson annulation is a two-step process that combines a Michael reaction with an intramolecular aldol reaction. It takes place between a nucleophilic donor, such as a β-keto ester; an enamine, or a β-diketone; and an α,β-unsaturated ketone acceptor, such as 3-buten-2-one. The product is a substituted 2-cyclohexenone.



3-Buten-2-one **Ethyl acetoacetate** **Michael product** **Annulation product**

The first step of the Robinson annulation is simply a Michael reaction. An enamine or an enolate ion from a β-keto ester or β-diketone effects a conjugate addition to an α,β-unsaturated ketone, yielding a 1,5-diketone. But as we saw in **Section 23.6**, 1,5-diketones undergo intramolecular aldol condensation to yield cyclohexenones when treated with base. Thus, the final product contains a six-membered ring, and an annulation has been accomplished. One example of this occurs during a synthesis of the steroid hormone estrone (**FIGURE 23.9**).

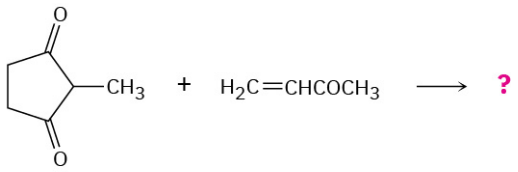


Michael acceptor (an α,β-unsaturated ketone) **Michael donor (a β-diketone)** **Michael product** **Robinson annulation product** **Estrone**

FIGURE 23.9 Synthesis of the steroid hormone estrone using a Robinson annulation reaction. The nucleophilic donor is a β-diketone.

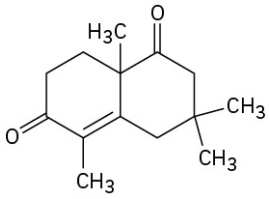
In this example, the β-diketone 2-methyl-1,3-cyclopentanedione is used to generate the enolate ion required for Michael reaction and an aryl-substituted α,β-unsaturated ketone is used as the acceptor. Base-catalyzed Michael reaction between the two partners yields an intermediate triketone, which then cyclizes in an intramolecular aldol condensation to give a Robinson annulation product. Several further transformations are required to complete the synthesis of estrone.

PROBLEM 23-21 What product would you expect from a Robinson annulation reaction of 2-methyl-1,3-cyclopentanedione with 3-buten-2-one?



2-Methyl-1,3-cyclopentanedione **3-Buten-2-one**

PROBLEM 23-22 How would you prepare the following compound using a Robinson annulation reaction between a β-diketone and an α,β-unsaturated ketone? Draw the structures of both reactants and the intermediate Michael addition product.



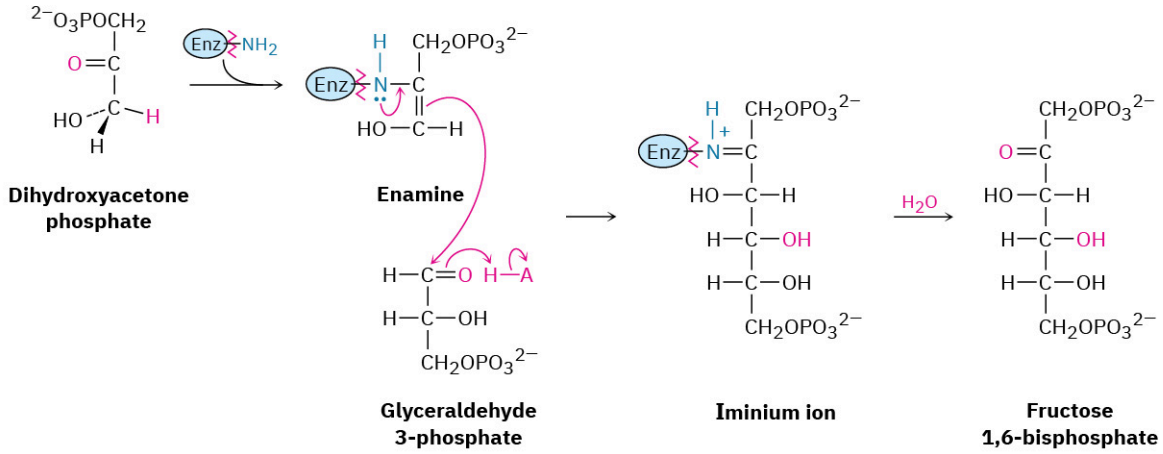
23.13 Some Biological Carbonyl Condensation Reactions

Biological Aldol Reactions

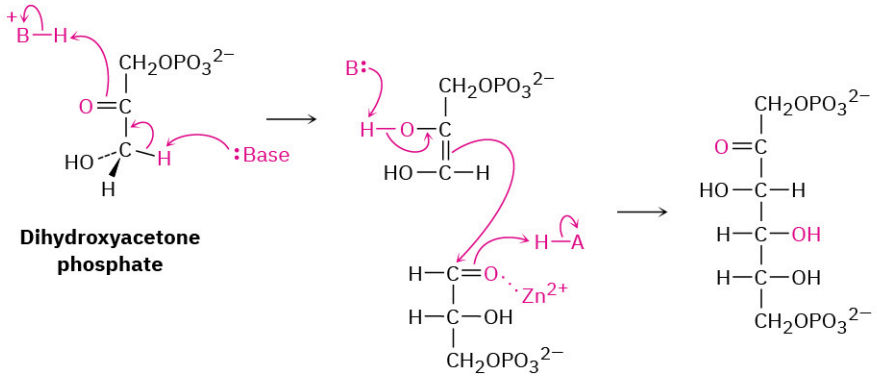
Aldol reactions occur in many biological pathways but are particularly common in carbohydrate metabolism, where enzymes called *aldolases* catalyze the addition of a ketone enolate ion to an aldehyde. Aldolases occur in all organisms and are of two types. Type I aldolases occur primarily in animals and higher plants; type II aldolases occur primarily in fungi and bacteria. Both types catalyze the same kind of reaction, but type I aldolases operate through an enamine while type II aldolases require a metal ion (usually Zn^{2+}) as a Lewis acid and operate through an enolate ion.

An example of an aldolase-catalyzed reaction occurs in glucose biosynthesis when dihydroxyacetone phosphate reacts with glyceraldehyde 3-phosphate to give fructose 1,6-bisphosphate. In animals and higher plants, dihydroxyacetone phosphate is first converted into an enamine by reaction with the $-NH_2$ group on a lysine amino acid in the enzyme. The enamine then adds to glyceraldehyde 3-phosphate, and the iminium ion that results is hydrolyzed. In bacteria and fungi, the aldol reaction occurs directly, with the ketone carbonyl group of glyceraldehyde 3-phosphate complexed to a Zn^{2+} ion to make it a better acceptor.

Type I aldolase



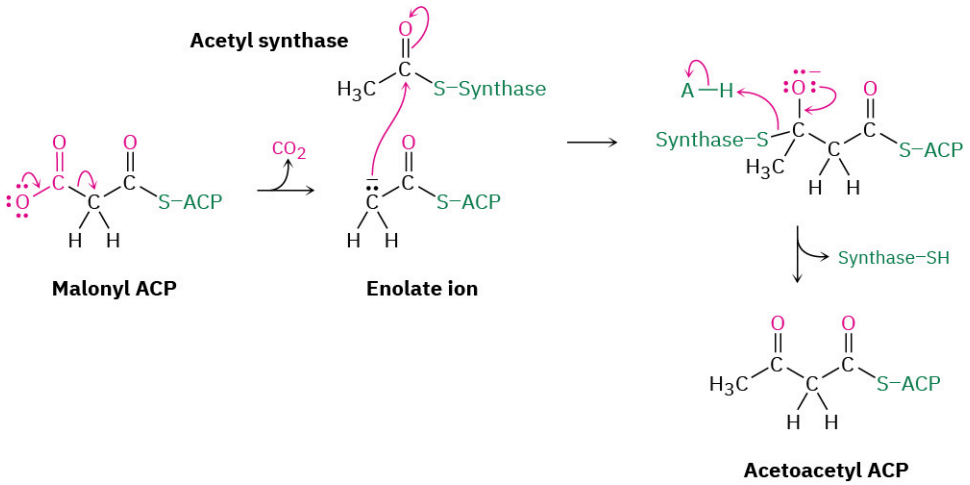
Type II aldolase



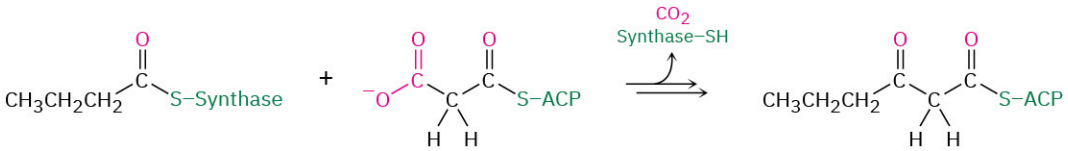
Note that the aldolase-catalyzed reactions are mixed aldol reactions, which take place between two different partners, as opposed to the symmetrical aldol reactions between identical partners usually carried out in the laboratory. Mixed aldol reactions often give mixtures of products in the laboratory but are successful in living systems because of the selectivity of the enzyme catalysts.

Biological Claisen Condensations

Claisen condensations, like aldol reactions, also occur in a large number of biological pathways. In fatty-acid biosynthesis, for instance, an enolate ion generated by decarboxylation (**Section 22.7**) of malonyl ACP adds to the carbonyl group of another acyl group bonded through a thioester linkage to a synthase enzyme. The tetrahedral intermediate that results then expels the synthase, giving acetoacetyl ACP. (The abbreviation ACP stands for acyl carrier protein, which forms thioester bonds to acyl groups.)



Mixed Claisen condensations also occur frequently in living organisms, particularly in the pathway for fatty-acid biosynthesis that we'll discuss in **Section 29.4**. Butyryl synthase, for instance, reacts with malonyl ACP in a mixed Claisen condensation to give 3-ketohexanoyl ACP.



CHEMISTRY MATTERS

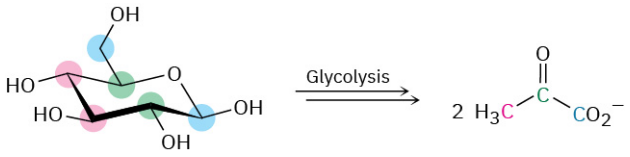
A Prologue to Metabolism

Biochemistry is carbonyl chemistry. Almost all metabolic pathways used by living organisms involve one or more of the four fundamental carbonyl-group reactions we've seen in the chapters on **Aldehydes and Ketones: Nucleophilic Addition Reactions** through **Carbonyl Condensation Reactions**. The digestion and metabolic breakdown of all the major classes of food molecules—fats, carbohydrates, and proteins—take place by nucleophilic addition reactions, nucleophilic acyl substitutions, α substitutions, and carbonyl condensations. Similarly, hormones and other crucial biological molecules are built up from smaller precursors by these same carbonyl-group reactions.



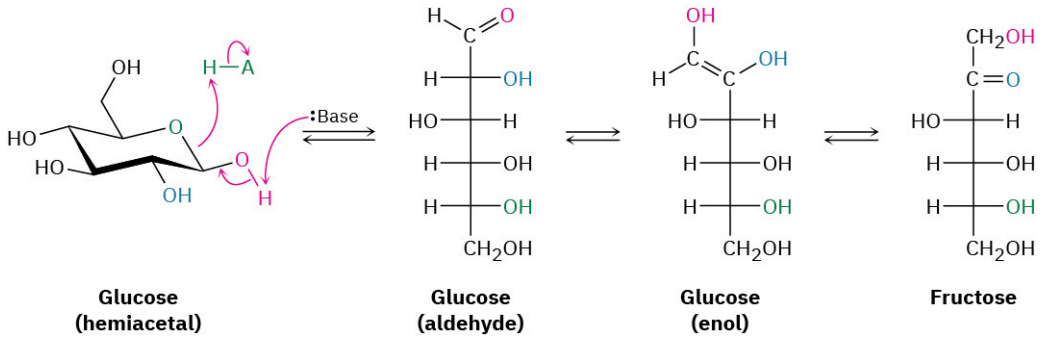
FIGURE 23.10 You are what you eat. Food molecules are metabolized by pathways that involve the four major carbonyl-group reactions. (credit: "Spring" by Giuseppe Arcimboldo, Louvre Museum/Wikimedia Commons, Public Domain)

Take *glycolysis,* for example, the metabolic pathway by which organisms convert glucose to pyruvate as the first step in extracting energy from carbohydrates.

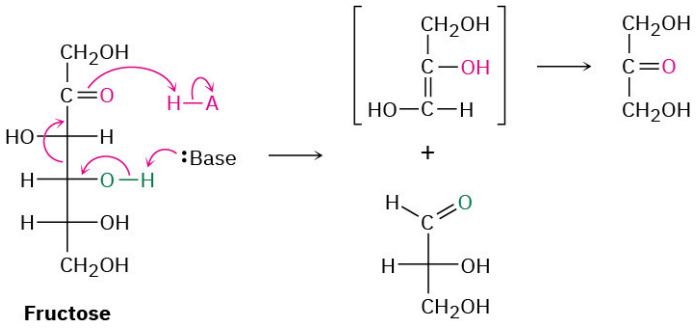


Glucose **Pyruvate**

Glycolysis is a ten-step process that begins with isomerization of glucose from its cyclic hemiacetal form to its open-chain aldehyde form—the reverse of a nucleophilic addition reaction. The aldehyde then undergoes tautomerization to yield an enol, which undergoes yet another tautomerization to give the ketone fructose.



Fructose, a β-hydroxy ketone, is then cleaved by a retro-aldol reaction into two three-carbon molecules—one ketone and one aldehyde. Further carbonyl-group reactions then occur until pyruvate is formed.



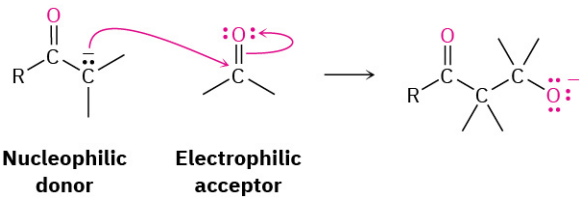
These few examples are only an introduction; we'll look at several of the major metabolic pathways in more detail in Chapter 29. The bottom line, however, is that you haven't seen the end of carbonyl-group chemistry if you have any interest in biological or medical fields. A solid grasp of carbonyl-group reactions is crucial to an understanding of biochemistry.

Key Terms

- **aldol reaction**
- **carbonyl condensation reaction**
- **Claisen condensation reaction**
- **Dieckmann cyclization**
- **Michael reaction**
- **Robinson annulation reaction**
- **Stork enamine reaction**

Summary

In this chapter, we've discussed the fourth and last of the common carbonyl-group reactions—the carbonyl condensation. A **carbonyl condensation reaction** takes place between two carbonyl partners and involves both nucleophilic addition and α-substitution processes. One carbonyl partner is converted by base into a nucleophilic enolate ion, which then adds to the electrophilic carbonyl group of the second partner. The first partner thus undergoes an α substitution, while the second undergoes a nucleophilic addition.



The **aldol reaction** is a carbonyl condensation that occurs between two aldehyde or ketone molecules. Aldol reactions are reversible, leading first to β-hydroxy aldehydes/ketones and then to α,β-unsaturated products after dehydration. Mixed aldol condensations between two different aldehydes or ketones generally give a mixture of all four possible products. A mixed reaction can be successful, however, if one of the two partners is an unusually good donor (ethyl acetoacetate, for instance) or if it can act only as an acceptor (formaldehyde and benzaldehyde, for instance). Intramolecular aldol condensations of 1,4- and 1,5-diketones are also successful and provide a good way to make five- and six-membered rings.

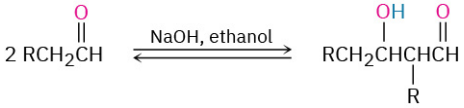
The **Claisen condensation reaction** is a carbonyl condensation that occurs between two ester components and gives a β-keto ester product. Mixed Claisen condensations between two different esters are successful only when one of the two partners has no acidic α hydrogens (ethyl benzoate and ethyl formate, for instance) and thus can function only as the acceptor partner. Intramolecular Claisen condensations, called **Dieckmann cyclization reactions**, yield five- and six-membered cyclic β-keto esters starting from 1,6- and 1,7-diesters.

The conjugate addition of a carbon nucleophile to an α,β-unsaturated acceptor is known as the **Michael reaction**. The best Michael reactions take place between relatively acidic donors (β-keto esters or β-diketones) and unhindered α,β-unsaturated acceptors. Enamines, prepared by reaction of a ketone with a disubstituted amine, are also good Michael donors in the Stork enamine reaction.

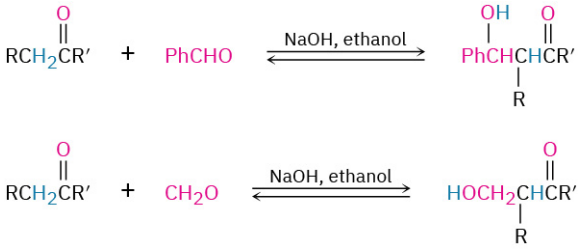
Carbonyl condensation reactions are widely used in synthesis. One example of their versatility is the **Robinson annulation reaction**, which leads to the formation of a substituted cyclohexenone. Treatment of a β-diketone or β-keto ester with an α,β-unsaturated ketone leads first to a Michael addition, which is followed by intramolecular aldol cyclization. Condensation reactions are also used widely in nature for the biosynthesis of such molecules as fats and steroids.

Summary of Reactions

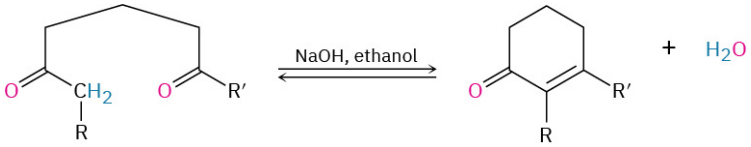
1. Aldol reaction (Section 23.1)



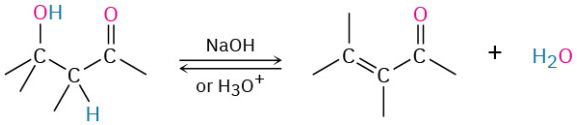
2. Mixed aldol reaction (Section 23.5)



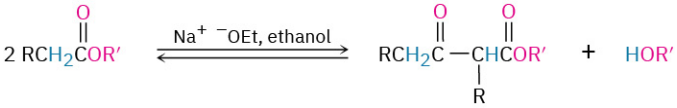
3. Intramolecular aldol reaction (Section 23.6)



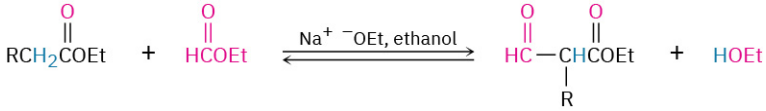
4. Dehydration of aldol products (Section 23.3)



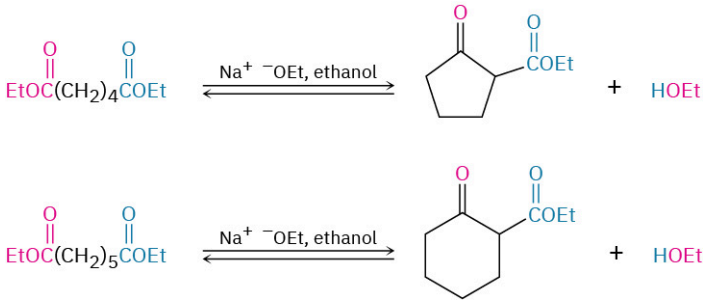
5. Claisen condensation reaction (Section 23.7)



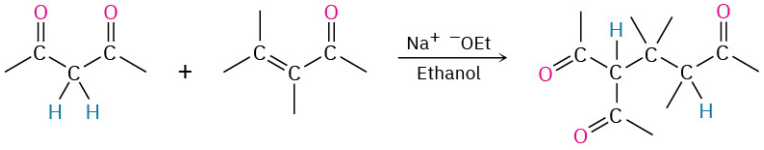
6. Mixed Claisen condensation reaction (Section 23.8)



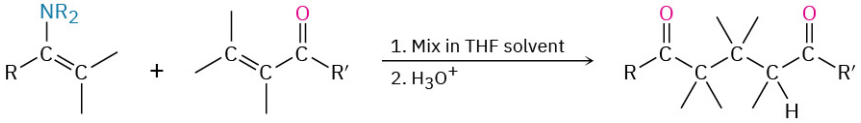
7. Intramolecular Claisen condensation (Dieckmann cyclization; Section 23.9)



8. Michael reaction (**Section 23.10**)



9. Carbonyl condensations with enamines (Stork enamine reaction; **Section 23.11**)

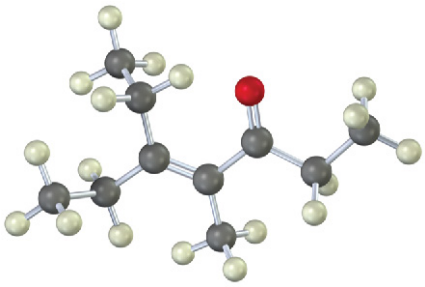


Additional Problems

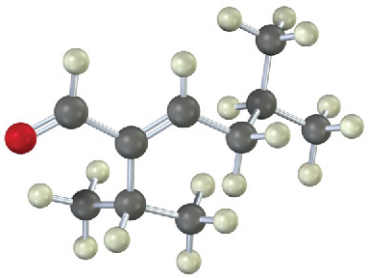
Visualizing Chemistry

PROBLEM 23-23 What ketones or aldehydes might the following enones have been prepared from by aldol reaction?

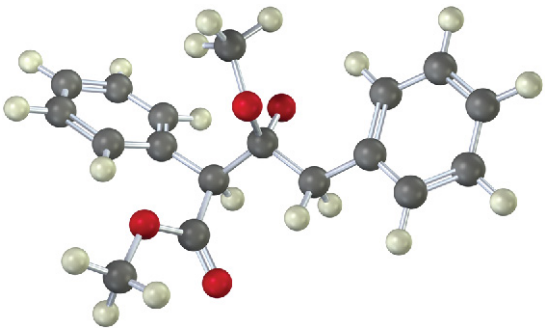
(a)



(b)



PROBLEM 23-24 The following structure represents an intermediate formed by addition of an ester enolate ion to a second ester molecule. Identify the reactant, the leaving group, and the product.



PROBLEM 23-25 The following molecule was formed by an intramolecular aldol reaction. What dicarbonyl precursor was used for its preparation?